Machin vs. Machine: An analysis of neural networks' ability to overcome unpredictability (Proposal)

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1 Introduction

1.1 Motivation

The human brain is an incredibly complicated example of biological machinery. It can store information and apply it to a variety of situations to efficiently solve problems, complete tasks, understand the world around them, and adapt to change. This allows humans to not only learn from a guide, but from their own mistakes and past experiences. Neural networks are artificial attempts to recreate this behaviour in machinery using calculus and matrices.

Currently, neural networks are believed to be helpful for image recognition, mathematical predictions, natural language processing, and playing games. I aim to build my own neural network in order to test these uses, and predict if it is possible for machinery to be as good at, or better, than humans at these common tasks in unpredictable environments.

1.2 Background

Neural networks consist of nodes that can receive multiple inputs , process, and output a real number, as can be seen in figure 1. As the machine learns, its outputs get closer and closer to a predicted output. This change in output is from changing the weight of each node. Each node always performs the same process with the sum of its inputs, but the influence that node has on the final output changes as the machine learns. The way the network decides the weight of a node is based on its learning paradigm. This research will focus on two paradigms: supervised learning and reinforcement learning.

1.2.1 Supervised Learning

This is the paradigm used for image recognition. Its input is a set of data with matching labels. During learning, the machine generalises the link between data and labels and changes the weights of nodes accordingly. Its applications include classification and regression.

1.2.2 Reinforcement Learning

This is the paradigm used for playing games. The machine interacts with its environment, attempting to maximise a reward that is controlled by an observer. At first, the interactions are random, but through trial and error the machine learns how to increase its reward and get closer to its goal. Its applications include training robots to complete physical activities, such as walking or grabbing items, and finding optimal ways to play games.

1.2.3 Backpropagation

Backpropagation is an algorithm that adjusts all weights across the network in order to reduce loss - the difference between its outputs and expected outputs. It uses chain rule to find the derivative of loss with respect to weight, and repeatedly adjusts the weight by a small amount in the opposite direction until the loss converges, which is where the loss is minimised. This is commonly used alongside supervised learning as it needs to

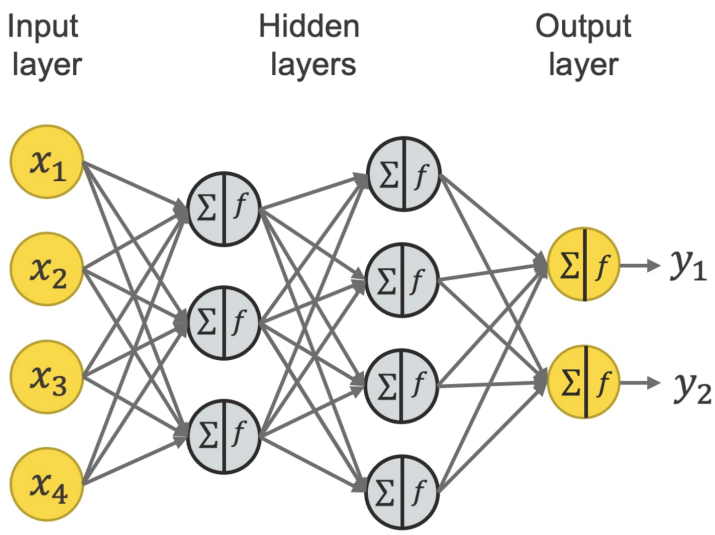

Figure 1: A diagram on a neural network with four layers of nodes. Σ represents the summation function and f represents the activation function. Taken from: [2]

have an expected output to function, which is provided in supervised learning but not reinforcement learning.

1.2.4 Activation function

This is the function that a neuron applies after the summation function - where the inputs are multiplied by their weights and summed together - to produce a single numerical output. It enables neural networks to learn non-linear relationships between data and labels. This is needed as weights produce linear relationships, but most real-world relationships are non-linear. Common examples of activation functions include: Sigmoid, Tanh, and ReLU [1].

2 Aims and Objectives

In this study I will measure the effectiveness of neural networks versus humans at facing brand new situations. I have broken this down into the following aims:

1. Create my own neural network
   (a) Program an image recognition network
   (b) Train the network on a dataset of labelled images
   (c) Test how the network reacts when faced with images containing objects it should recognise, as well as objects it should not
2. Test publicly available neural networks

(a) Test a selection of pre-trained image recognition models on the same dataset as the model I created
(b) Research how well neural networks have performed at other tasks, including, but not limited to, prediction activities and playing games

3. Test humans at the tasks completed by the neural networks in the above aims
4. Analyse the similarities and differences between the accomplishments of the neural networks and the humans at these tasks

3 Literature Review

3.1 Generalisation in humans and deep neural networks [3]

The authors of this paper aimed to compare the success rate of humans against deep neural networks when tasked with recognising distorted images. They found that humans were mostly more accurate on average at identifying distorted images than the networks for non-colour related distortions, where the networks always had a higher average accuracy. They also found that human participants' responses were distributed almost entirely equally, whereas the networks showed increasing biases towards certain categories as the distortions got stronger.

My research differs because, rather than focusing on how different types of distortion affect accuracy, I will focus on how unpredictability could affect the overall practicality of neural networks. This will take into account not only accuracy, but also time consumption, resource consumption, and the effect even just one outlier could have on the reliability of its results.

3.2 How Neural Networks Extrapolate: From Feedforward to Graph Neural Networks [4]

The authors of this paper aimed to find the conditions required for feedforward neural networks, aka. multilayer perceptrons (MLPs) to be able to extrapolate data well. They trained a MLP using the ReLU activation function on a variety of non-linear functions, but, when tested, the MLP always "linearised" the data almost immediately after leaving the domain of the test function, leading to linear extrapolations that do not align with the correct non-linear target function. This means that MLPs extrapolate best when the target function is linear.

My research differs from this because I am going to focus on what happens when back-propagated neural networks try to extrapolate data in normal conditions, as opposed to the feedforward neural networks researched in this article.

3.3 Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification [5]

The authors of this paper aimed to develop an activation function that could allow a neural network to surpass a human's ability to classify images. They altered the ReLU

activation function to now include parameters that could change as the network learns, allowing for more variance between each node of the network, and therefore increasing the network's ability to improve its output accuracy. This new activation function allowed for a neural network with an image classification error rate of 4.94%, which is lower than the average human's error rate of 5.1%.

My research differs from this because I will focus on how data it wasn't trained for affects a neural network's reliability, rather than improving the reliability of a neural network on data it was trained for.

4 Methodology

4.1 Aim 1: Creating a neural network

I will program my neural network in Python within Google Colab so that I can access Google's accelerated hardware to speed up the training process [6]. This will use supervised learning with the backpropagation training algorithm and ReLU activation function. I will use the CIFAR-10 dataset [7] to train and test the network, as well as some other images that are unrecognisable - due to heavy distortion - for both the network and any human in the testing. This aim is highly scalable based on time constraints as I could build the entire network from scratch, I could use Python libraries to to complete some functions, or I could use pre-trained networks online.

4.2 Aim 2: Public neural networks

I aim to get access to some of the leading neural networks in image recognition and test them against my edited CIFAR-10 testing dataset. I am especially interested by how they respond to the distorted images.

I will also research other applications of neural networks, and the current networks whose results point to them being great representatives for this application. I would like to gather data about them that includes the kind of data they were trained on, the kind of data they were tested on, and the results of those tests. I would also like to test them myself against edited datasets that contains unfamiliar data in order to see how different networks react to the unexpected.

4.3 Aim 3: Human testing

I will gather a sample of EMS students using opportunistic sampling to test their ability to complete the same tasks the neural networks completed. I will measure their accuracy, a score 1-10 of how certain they are, the time taken, and any comments they may have about the data - especially the unexpected data. I will use this data only as a baseline to compare the neural networks to rather than a direct comparison as the lack of reliability and large amount of bias that comes from such a small sample will affect the data too much for direct comparisons to be helpful towards this research.

4.4 Aim 4: Analyse results

I will use R find the similarities and differences between neural networks and humans at a variety of metrics. These metrics may include mean success rate as a decimal, variance in success across repeated tests, mean confidence rating as a decimal, mean time taken, and observations of responses to unexpected data. I will also greatly focus on how outliers affect the data, as one large outlier doesn't typically skew statistical data that much, but it can have a great impact on the real world, so I want to ensure this is properly taken into account in my analysis by singling out outliers I believe to be of greater importance than my statistical methods may represent them to be. I will then represent these metrics using suitable diagrams, such as histograms for relationships between time and another metric, or scatter plots with a line of regression for the relationship between a dependent and an independent metric, and analyse these results in context to give predictions about the future of machine learning.

5 Timeline

Task	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Program and train a neural network										
Research and test publicly available neural networks										
Test my network										
Test humans										
Analyse results										
Write up										

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